

HIGH-ACRYLONITRILE NBR PATENT LANDSCAPE: SYNTHESIS, DEGRADATION, AND FORMULATION-DRIVEN PROCESSING ROUTES

1. ABSTRACT

This report surveys patents relevant to high-acrylonitrile nitrile butadiene rubber (NBR), with emphasis on how acrylonitrile-rich nitrile elastomers are prepared, modified, or incorporated into downstream formulations. The target material is defined as NBR with elevated acrylonitrile content relative to standard grades, typically requiring careful control of butadiene/acrylonitrile feed composition, polymerization route, and post-polymerization processing to preserve desired polarity, oil resistance, and processability. The patent set spans three distinct technical directions: direct emulsion polymerization of acrylonitrile/butadiene copolymers, post-polymerization metathesis degradation of nitrile rubber to lower molecular weight or alter architecture, and formulation/composite patents that use NBR or HNBR as a functional elastomeric component.

The most directly relevant synthesis disclosure is the emulsion polymerization route described in US7592409B2, which explicitly states that the B block can comprise acrylonitrile and a conjugated diolefin such as 1,3-butadiene, and that controlled emulsion polymerization can be used to make block architectures incorporating nitrile-rubber segments. EP2057205B1 and EP2473279B1 are not synthesis patents for high-ACN NBR per se, but they are highly relevant because they show how nitrile rubbers are transformed by metathesis, including use of Grubbs II catalyst and hydrogenation with Wilkinson's catalyst, enabling molecular-weight reduction and property tuning. EP3327047A1 demonstrates RAFT-based controlled polymerization in emulsion and explicitly includes acrylonitrile among the monomer classes, indicating a broader controlled-polymerization toolbox that could be adapted to nitrile-containing systems. The remaining patents are formulation/process patents: EP2561011B1 uses butadiene-acrylonitrile copolymer as a nitrile resin base in conductive composites, US9909003B2 uses carboxyl-terminated butadiene acrylonitrile rubber in fire-retardant epoxy prepregs, and US10716877B2 uses NBR in nanocellulose-reinforced elastomer composites.

Across the landscape, the dominant trends are emulsion-based synthesis, controlled radical polymerization, metathesis-based molecular-weight adjustment, and heavy emphasis on downstream compounding with fillers, coagents, and reinforcing phases. For high-acrylonitrile

NBR synthesis, the evidence supports monomer-ratio control in emulsion polymerization as the most direct route, while metathesis and formulation patents provide useful process/property context rather than primary synthesis recipes.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473279B1
- Patent Title: Metathesis of a nitrile rubber in the presence of a transition metal catalyst
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis degradation of nitrile rubber in solution using a transition-metal catalyst, followed by hydrogenation to obtain hydrogenated nitrile rubber with reduced molecular weight and altered processability. The examples use monochlorobenzene as solvent, a metathesis catalyst, and then hydrogenation with tris(triphenylphosphine)rhodium chloride under hydrogen pressure. - Nitrile rubber charge: 700 g — Examples 5-6 - Solvent: 4667 g monochlorobenzene — Examples 5-6 - Reactor volume: 10 L — Examples 5-6 - 1-Hexene addition: 4 phr — Examples 5-6 - Metathesis reaction time: 2 h — Examples 5-6 - Metathesis temperature: 22 °C — Examples 5-6 - Hydrogenation catalyst: tris-(triphenylphosphine) rhodium chloride 0.06 phr — Examples 5-6 - Hydrogen pressure: 85 bar — Examples 5-6 - Hydrogenation time: 4 h — Examples 5-6 - Hydrogenation temperature: 138 °C — Examples 5-6 - Agitation speed: 600 rpm — Examples 5-6 - Example 5 metathesis catalyst: 0.07 phr — Example 5 - Example 5 solvent content: 200 ppm — Example 5 - Example 5 Mn: 9,300 g/mol — Example 5 - Example 5 Mw: 13,700 g/mol — Example 5 - Example 5 PDI: 1.48 — Example 5 - Example 5 hydrogenated Mn: 8,500 g/mol — Example 5 - Example 5 hydrogenated Mw: 12,000 g/mol — Example 5 - Example 5 hydrogenated PDI: 1.42 — Example 5 - Example 6 metathesis catalyst: 0.01 phr — Example 6 - Example 6 solvent content: 1,900 ppm — Example 6 - Example 6 Mn: 12,700 g/mol — Example 6 - Example 6 Mw: 21,000 g/mol — Example 6 - Example 6 PDI: 1.67 — Example 6 - Example 6 hydrogenated Mn: 11,000 g/mol — Example 6 - Example 6 hydrogenated Mw: 17,500 g/mol — Example 6 - Example 6 hydrogenated PDI: 1.58 — Example 6 - Example 7 metathesis catalyst: 0.1 phr — Table 3 - Example 7 solvent content: 1,300 ppm — Table 3 - Example 7 Mn: 19,000 g/mol — Table 3 - Example 7 Mw: 34,000 g/mol — Table 3 - Example 7 PDI: 1.78 — Table 3

Relevant Experimental Evidence - Example 5: low catalyst loading produced lower-Mw HNBR after metathesis and hydrogenation. - Example 6: even lower catalyst loading gave higher Mn and Mw than Example 5, showing catalyst level affects degradation extent. - Example 7: changing the nitrile rubber feedstock changed the resulting molecular-weight profile after the same metathesis/hydrogenation sequence. - Examples 8-36: the low-viscosity HNBR was compounded with fillers, coagents, and peroxide cure systems to demonstrate processability and vulcanizate performance. - Examples 56-60: blending the low-viscosity HNBR reduced compound Mooney viscosity and improved injection-molding fill behavior without added plasticizer.

Technical Relevance This patent is relevant because it shows how nitrile rubber molecular weight and processability can be tuned by metathesis and hydrogenation, which is important when handling acrylonitrile-rich NBR feedstocks or derivatives. Although it does not create high-ACN NBR directly, it provides a practical route for modifying nitrile rubber properties after synthesis.

Patent 2

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses controlled emulsion polymerization using in-situ emulsifier formation and controlled radical polymerization agents. The core disclosure explicitly includes block polymers where the B block contains acrylonitrile or methacrylonitrile together with a conjugated diolefin such as 1,3-butadiene, making it directly relevant to nitrile-rubber-type block architectures. - Styrene charge: 1,000 g — Example 2 - Oleic acid: 60.0 g — Example 2 - Dibenzyltrithiocarbonate: 7.2 g — Example 2 - RO water: 4,000 g — Example 2 - Potassium persulfate: 40.0 g — Example 2 - Tripotassium phosphate: 40.0 g — Example 2 - Potassium hydroxide: 16.4 g — Example 2 - Reaction temperature: 65 °C — Example 2 - Complete conversion time: less than 1.5 h — Example 2 - Styrene charge: 1,000 g — Example 3 - Hexadecyl half ester of maleic acid: 60.0 g — Example 3 - Dibenzyltrithiocarbonate: 7.2 g — Example 3 - RO water: 4,000 g — Example 3 - Potassium persulfate: 40.0 g — Example 3 - Tripotassium phosphate: 40.0 g — Example 3 - Potassium hydroxide: 16.4 g — Example 3 - Reaction temperature: 65 °C — Example 3 - Complete conversion time: less than 3 h — Example 3 - Styrene charge: 1,000 g — Example 6 - Oleic acid: 60.0 g — Example 6 - PBN: 10.0 g — Example 6 - RO water: 4,000 g — Example 6 - Potassium persulfate: 40.0 g — Example 6 - Tripotassium phosphate: 40.0 g — Example 6 - Potassium hydroxide: 16.4 g — Example 6 - Reaction temperature: 75 °C — Example 6 - Complete conversion time: less than 2 h — Example 6 - Styrene charge: 50 g (0.48 moles) — Example 8 - Oleic acid: 3.0 g (0.0106 moles) — Example 8 - PBN: 0.44 g (0.002485 moles) — Example 8 - Distilled water: 200 g — Example 8 - Potassium

persulfate: 2.06 g (0.00762 moles) — Example 8 - Tripotassium phosphate: 2.13 g (0.01 moles) — Example 8 - Potassium hydroxide: 0.9 g (0.014 moles) — Example 8 - Reaction temperature: 65 °C — Example 8 - Complete conversion time: about 7 h — Example 8 - Styrene charge: 200 g (1.923 moles) — Example 10 - Oleic acid: 12.0 g (0.0424 moles) — Example 10 - PBN: 2.0 g (0.0113 moles) — Example 10 - Distilled water: 500 g — Example 10 - Potassium persulfate: 8.0 g (0.0296 moles) — Example 10 - Tripotassium phosphate: 8.0 g (0.0377 moles) — Example 10 - Potassium hydroxide: 3.6 g (0.056 moles) — Example 10 - Reaction temperature: 75 °C — Example 10 - Complete conversion time: about 3.5 h — Example 10

Relevant Experimental Evidence - Example 2: RAFT emulsion polymerization of styrene achieved complete conversion in less than 1.5 h at 65 °C. - Example 3: in-situ generated surfactant enabled rapid microemulsion formation and complete conversion in less than 3 h. - Example 6: NMP with PBN at 75 °C gave a stable latex and lower Mn than the 65 °C case. - Example 8: small-scale NMP emulsion polymerization at 65 °C produced controlled Mn and PDI. - Example 10: higher-solids NMP at 75 °C still gave controlled molecular weight and low PDI.

Technical Relevance This patent is directly relevant because it explicitly claims block polymers containing acrylonitrile or methacrylonitrile with isoprene and describes controlled emulsion polymerization methods that can be adapted to nitrile-rich copolymers. It provides a process framework for composition control, surfactant formation, and controlled radical polymerization in aqueous media.

Patent 3

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis-driven segment interchange between ethylenically unsaturated polymers to form meta-block copolymers. In the examples, partially hydrogenated nitrile/butadiene rubber is dissolved with another polymer in solvent and treated with Grubbs II catalyst, then isolated by precipitation and filtration. - Polymer charge: 0.25 g each of HPBD and HNBR — Example 1 - Solvent: toluene — Example 1 - Dissolution temperature: 95 °C — Example 1 - Catalyst: 8 mg Grubbs II metathesis catalyst — Example 1 - Reaction time: 1 h — Example 1 - Precipitation solvent: methanol — Example 1 - Polymer charge: 0.25 g each of unsaturated polyurethane and poly(ethylene-co-butadiene) — Example 2 - Solvent: tetrachloroethane — Example 2 - Dissolution temperature: 105 °C — Example 2 - Catalyst: 9 mg Grubbs II metathesis catalyst — Example 2 - Reaction time: 1 h — Example 2 - Extraction solvent: 30 ml tetrahydrofuran — Example 2 - Polymer charge: 1.05 g

fumaryl-modified polycarbonate and 0.45 g poly(ethylene-co-butadiene) — Example 3 - Solvent: toluene — Example 3 - Dissolution temperature: 105 °C — Example 3 - Catalyst: 9 mg Grubbs II metathesis catalyst — Example 3 - Reaction time: 1 h — Example 3 - Polymer charge: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-butadiene) — Example 4 - Solvent: toluene — Example 4 - Dissolution temperature: 105 °C — Example 4 - Catalyst: 9 mg Grubbs II metathesis catalyst — Example 4 - Reaction time: 1 h — Example 4 - Polymer charge: 0.25 g each of unsaturated poly(ethylene oxide) and poly(ethylene-co-butadiene) — Example 5 - Solvent: toluene — Example 5 - Dissolution temperature: 105 °C — Example 5 - Catalyst: 9 mg Grubbs II catalyst — Example 5 - Reaction time: 1 h — Example 5 - Polymer charge: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-octene-co-butadiene) — Example 6 - Solvent: toluene — Example 6 - Dissolution temperature: 105 °C — Example 6 - Catalyst: 9 mg Grubbs II metathesis catalyst — Example 6 - Reaction time: 1 h — Example 6 - Polymer charge: 0.10 g poly(ethylene-co-butadiene) and 0.40 g poly(ethylene-co-octene-co-butadiene) — Example 7 - Solvent: toluene — Example 7 - Dissolution temperature: 105 °C — Example 7 - WCl₆ catalyst: 8,000 ppm — Example 7 - Tri-n-butylmethyltin co-catalyst: 13,000 ppm — Example 7 - Reaction time: 1 h — Example 7 - Polymer charge: 0.25 g each of unsaturated polyester and poly(ethylene-co-butadiene) — Example 8 - Solvent: toluene — Example 8 - Dissolution temperature: 105 °C — Example 8 - Catalyst: 9 mg Grubbs II catalyst — Example 8 - Reaction time: 1 h — Example 8 - Polymer charge: 0.25 g SBS and 0.25 g poly(ethylene-co-butadiene) — Example 9 - Solvent: toluene — Example 9 - Dissolution temperature: 105 °C — Example 9 - WCl₆ catalyst: 8,000 ppm — Example 9 - Tri-n-butylmethyltin co-catalyst: 13,000 ppm — Example 9 - Reaction time: 1 h — Example 9 - Polymer charge: 0.25 g unsaturated fumaryl-modified polycarbonate, 0.25 g SBS, and 0.25 g poly(ethylene-co-butadiene) — Example 10 - Solvent: toluene — Example 10 - Dissolution temperature: 105 °C — Example 10 - Catalyst: 9 mg Grubbs II catalyst — Example 10 - Reaction time: 1 h — Example 10 - Extraction solvent: 50 ml tetrahydrofuran — Example 10

Relevant Experimental Evidence - Example 1: HNBR-containing blend formed a meta-block copolymer, while the no-catalyst control remained a simple blend. - Example 2: segment interchange between polyurethane and polyethylene-co-butadiene was confirmed by FTIR after metathesis. - Example 3: TEM and SAXS showed ordered microphase morphology consistent with meta-block formation. - Example 7: tungsten hexachloride/organotin catalysis produced a meta-block copolymer with altered crystallization behavior. - Example 10: a three-component metathesis system yielded a terpolymer-like meta-block structure confirmed by NMR, TEM, and SAXS.

Technical Relevance This patent is relevant because it explicitly includes HNBR as a reagent polymer and demonstrates how nitrile-rubber-containing materials can be reorganized into meta-block architectures by metathesis. It is useful for understanding post-polymerization modification of nitrile elastomers rather than direct high-ACN synthesis.

Patent 4

- Patent Number: EP2561011B1

- Patent Title: Thermoplastic and/or elastomeric composite material based on carbon nanotubes ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of agglomerated carbon nanotube/graphene/polymer composite materials by co-kneading and subsequent dilution into thermoplastic or elastomeric matrices. One example specifically uses a butadiene-acrylonitrile copolymer as a nitrile resin base, preheated and injected into a BUSS co-kneader with nanotubes, graphenes, and an acrylic copolymer. - Co-kneader model: BUSS MDK 46 — Example 7 - L/D: 11 — Example 7 - Nitrile resin preheat temperature: 160 °C — Example 7 - Injection temperature: 190 °C — Example 7 - Co-kneader temperature: 200 °C — Example 7 - Flow rate: 12 kg/h — Example 7 - Screw speed: 240 rpm — Example 7 - Composite composition: 35% by weight CNT — Example 7 - Composite composition: 5% by weight graphenes — Example 7 - Composite composition: 55% by weight nitrile resin — Example 7 - Composite composition: 5% by weight acrylic copolymer — Example 7 - Co-kneader model: BUSS MDK 46 — Example 8 - L/D: 11 — Example 8 - Kneading temperature: 90-110 °C — Example 8 - Composite composition: 30% by weight CNT — Example 8 - Composite composition: 5% by weight graphenes — Example 8 - Composite composition: 65% by weight silicone resin — Example 8 - Kneading temperature: 160-180 °C — Example 9 - Composite composition: 30% by weight CNT — Example 9 - Composite composition: 5% by weight graphenes — Example 9 - Composite composition: 64% by weight fluoroelastomer — Example 9 - Composite composition: 1% by weight processing aid — Example 9 - Kneading temperature: 160-180 °C — Example 10 - Composite composition: 35% by weight CNT — Example 10 - Composite composition: 5% by weight graphenes — Example 10 - Composite composition: 39% by weight fluoroelastomer — Example 10 - Composite composition: 20% by weight liquid fluoroelastomer resin — Example 10 - Composite composition: 1% by weight processing aid — Example 10 - Co-kneader model: BUSS MDK 46 — Example 11 - UD: 11 — Example 11 - Zone 1 temperature: 150 °C — Example 11 - Zone 2 temperature: 140 °C — Example 11 - Flow rate: 12 kg/h — Example 11 - Screw speed: 200 rpm — Example 11 - Composite composition: 18% by weight CNT — Example 11 - Composite composition: 2% by weight graphenes — Example 11

Relevant Experimental Evidence - Example 7: a nitrile-resin-based composite granulate containing CNT, graphenes, nitrile resin, and acrylic copolymer was produced. - Example 8: silicone elastomer-based composite granules were prepared at 90-110 °C. - Example 9: fluorinated elastomer composite granules were produced at 160-180 °C. - Example 10: a higher CNT loading with liquid fluoroelastomer resin was processed into granules. - Example 11: solid fluoroelastomer feed with CNT and graphenes was processed at 150/140 °C and 12 kg/h.

Technical Relevance This patent is relevant because it explicitly uses a butadiene-acrylonitrile copolymer as a nitrile resin base in a composite processing route. While not a synthesis patent for high-ACN NBR, it shows how nitrile elastomers are compounded and processed in high-filler systems.

Patent 5

- Patent Number: EP3327047A1
- Patent Title: Thermoplastic elastomers via reversible addition-fragmentation chain transfer ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses RAFT polymerization in emulsion and solution, including block and statistical copolymers from monomers that can include acrylonitrile. The examples focus on styrene and acrylated epoxidized soybean/corn oil systems, but the core claims and synthesis description establish a controlled radical polymerization platform that explicitly encompasses nitrile monomers. - Monomer: styrene — Example 1 - Initiator: AIBN — Example 1 - Chain transfer agent: 1-phenylethyl benzodithioate — Example 1 - Monomer: solvent ratio: various mass ratios — Example 1 - Initiator:CTA ratio: 1:5 molar ratio — Example 1 - Monomer:CTA ratio: 10:1 molar ratio — Example 1 - Reaction temperature: 100 °C — Example 1 - Reaction temperature: 70 °C — Example 1 - Reaction temperature: 100 °C — Example 4 - Reaction temperature: 70 °C — Example 4 - CuIICl₂ addition: 0.1% by mass relative to polymer — Example 6

Relevant Experimental Evidence - Example 1: RAFT synthesis of styrene and AESO-derived polymers was demonstrated with controlled molecular weight. - Example 2: characterization showed block copolymers with defined styrene content and PDI. - Example 4: telechelic CTA enabled triblock synthesis and post-reaction purification. - Example 5: statistical copolymers from styrene and AESO/AECO were prepared by simultaneous RAFT polymerization. - Example 6: post-polymerization CuCl₂ treatment modified chain ends to improve compatibility.

Technical Relevance This patent is relevant because its claims explicitly include acrylonitrile monomers within a controlled RAFT polymerization framework. Although the examples do not use acrylonitrile, the disclosed process is a plausible route for controlled synthesis of acrylonitrile-rich copolymers or block copolymers.

Patent 6

- Patent Number: US9909003B2
- Patent Title: Fire-retardant composite materials

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses fire-retardant epoxy prepregs and composite materials that incorporate toughening additives, including carboxyl-terminated butadiene acrylonitrile rubber, along with mineral fillers and curing catalysts. The nitrile rubber is used as a reactive toughener in an epoxy matrix rather than as a synthesized NBR product. - Difunctional-terminated rubber: 1 to 15 wt % of resin matrix — Example a - Polyhydroxyl ether: 2 to 15 wt % of resin matrix — Example a - Rubber dispersion: 3 to 15 wt % of resin matrix — Example a - Fire-retardant mineral filler: 15 to 30 wt % of resin matrix — Example a - Smoke-suppressant mineral filler: 1 to 10 wt % of resin matrix — Example a - Matrix resin composition: 31 to 38 wt % non-halogenated bisphenol F epoxide resin — Example 1 - Phosphorus content: 3 to 5 wt % added phosphorus — Example 1 - Tri-functional epoxide diluent: 5 to 15 wt % — Example 1 - Catalyst carrier: up to 10 wt % — Example 1 - Fire test method: ISO 5660-1 — Example 1 - Fire test method: UL94 — Example 1 - Burn time for phosphorus-modified epoxide: 0-3 s — Example 1 - Time to ignition: 37 s — Example 1 - Peak rate of heat release: 177 kWm⁻² — Example 1 - Total heat released: 25 MJm⁻² — Example 1 - Smoke emission: 234 MWkg⁻¹ — Example 1 - Burn time with mineral filler: 0 s — Example 1 - Time to ignition with mineral filler: 46 s — Example 1 - Peak rate of heat release with mineral filler: 170 kWm⁻² — Example 1 - Total heat released with mineral filler: 24 MJm⁻² — Example 1 - Smoke emission with mineral filler: 107 MWkg⁻¹ — Example 1 - Burn time with imidazole curing agents: 0 s — Example 1 - Time to ignition with imidazole curing agents: 51 s — Example 1 - Peak rate of heat release with imidazole curing agents: 175 kWm⁻² — Example 1 - Total heat released with imidazole curing agents: 23 MJm⁻² — Example 1 - Smoke emission with imidazole curing agents: 105 MWkg⁻¹ — Example 1

Relevant Experimental Evidence - Example a: the patent identifies carboxyl-terminated butadiene acrylonitrile rubber as a toughening additive in the resin matrix. - Example 1: phosphorus modification improved fire performance relative to unmodified epoxides. - Example 1: adding mineral filler further reduced smoke emission and improved fire-retardant behavior. - Example 1: imidazole curing agents reduced cure temperature/time while maintaining improved FST performance.

Technical Relevance This patent is relevant because it uses butadiene acrylonitrile rubber in a high-performance composite formulation, showing compatibility of nitrile elastomers with reactive epoxy systems. It does not synthesize high-ACN NBR, but it is useful for downstream application and formulation context.

Patent 7

- Patent Number: US10716877B2
- Patent Title: Nanocomposite elastomers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nanocomposite elastomers prepared by dispersing nanocellulose into elastomer matrices, including a nitrile butadiene rubber example. The NBR example uses solid NBR dissolved in DMF, followed by addition of nanocellulose dispersion, casting, and curing under nitrogen. - Nanocellulose loading: 6.5 wt % of dry rubber mass — Example 4 - NBR charge: 5 g — Example 4 - DMF volume: 100 mL — Example 4 - NFC dispersion: 10 mL of 3.5 w/v % NFC in DMF — Example 4 - Stirring: overnight at room temperature — Example 4 - Curing temperature: 45 °C — Example 4 - Curing time: 48 h — Example 4 - NaOH treatment: 2% (w/v) sodium hydroxide solution — Example 5 - Treatment temperature: 80 °C — Example 5 - Treatment time: 2 h — Example 5 - Homogenization pressure: 700 bar — Example 5 - Bleaching: 1% (w/v) sodium chlorite — Example 6 - Bleaching temperature: 70 °C — Example 6 - Bleaching time: 1 h — Example 6 - Homogenization pressure: 700 bar — Example 6 - Carboxymethylation reagent: 2% (w/v) monochloroacetic acid — Example 7 - Carboxymethylation temperature: 80 °C — Example 7 - Carboxymethylation time: 1 h — Example 7 - Homogenization pressure: 700 bar — Example 7 - Choline chloride/urea ratio: 2:1 molar ratio — Example 8 - Treatment temperature: 100 °C — Example 8 - Treatment concentration: 3.7 wt % — Example 8 - Homogenization pressure: 700 bar — Example 8 - Sulfur loading: 0.6% wt of total solids mass — Example 3 - ZnO loading: 0.2% wt of total solids mass — Example 3 - ZDEC loading: 0.5% wt — Example 3 - Cure time: 20 min — Example 3 - Cure temperature: 125-130 °C — Example 3 - Latex solids content: 45 wt. % — Example 11 - Nanocellulose loading: 0.1 wt. % of latex solids content — Example 11 - Stirring speed: 50 rpm — Example 11 - Drying temperature: 50 °C — Example 11 - Final drying temperature: 125 °C — Example 11 - Final drying time: 25 min — Example 11 - Air burst pressure: 1.4 kPa — Example 11 - Air burst volume: 38.5 L — Example 11 - Stress at break: 27 MPa — Example 11 - Stress at 500% elongation: 1.9 MPa — Example 11 - Nanocellulose loading: 0.1 wt. % of latex solids content — Example 12 - Stirring speed: 50 rpm — Example 12 - Drying temperature: 50 °C — Example 12 - Final drying temperature: 125 °C — Example 12 - Final drying time: 25 min — Example 12 - Air burst pressure: 1.3 kPa — Example 12 - Air burst volume: 33 L — Example 12 - Stress at break: 27 MPa — Example 12 - Stress at 500% elongation: 0.7 MPa — Example 12 - Nanocellulose loading: 0.4 wt. % of latex solids content — Example 14 - Stirring speed: 50 rpm — Example 14 - Drying temperature: 50 °C — Example 14 - Final drying temperature: 125 °C — Example 14 - Final drying time: 25 min — Example 14 - Air burst pressure: 1.1 kPa — Example 14 - Air burst volume: 31.5 L — Example 14 - Vacuum drying temperature: 70 °C — Example 15 - Vacuum drying time: 24 h — Example 15 - CNC concentration: 1 wt % — Example 15 - Casting temperature: 60 °C — Example 15 - Annealing temperature: 70 °C — Example 15 - Annealing

time: 6 h — Example 15

Relevant Experimental Evidence - Example 4: NBR was dissolved in DMF and reinforced with nanocellulose to form a cast composite film. - Example 3: sulfur/ZnO/ZDEC curing of synthetic rubber with nanocellulose was demonstrated. - Examples 11-14: latex dipping produced nanocellulose-reinforced condoms with measured burst and tensile properties. - Example 15: solvent-cast TPU/nanocellulose films were prepared as a comparison elastomer composite system.

Technical Relevance This patent is relevant because it explicitly uses NBR in a nanocomposite elastomer formulation, showing how nitrile rubber can be processed with reinforcing nanocellulose. It does not address synthesis of high-ACN NBR, but it is useful for downstream property and processing context.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7592409B2 is the only patent in the set that explicitly frames acrylonitrile-containing polymer blocks as part of the synthesis claims; it uses controlled emulsion polymerization with in-situ emulsifier formation and RAFT/NMP/ATRP-compatible control chemistry.
- EP2473279B1 and EP2057205B1 are post-polymerization modification patents: both use metathesis chemistry on nitrile-rubber-containing substrates, but EP2473279B1 focuses on molecular-weight reduction/hydrogenation, whereas EP2057205B1 focuses on segment interchange to create meta-block copolymers.
- EP3327047A1 is a controlled radical polymerization patent with explicit acrylonitrile claim language, but its examples are styrene/AESO/AECO-based; it is broader in polymerization platform but less directly nitrile-rubber-specific than US7592409B2.
- The formulation patents (EP2561011B1, US9909003B2, US10716877B2) all use nitrile rubber or butadiene-acrylonitrile rubber as a matrix/toughener, but none disclose synthesis of high-ACN NBR; they mainly show compounding, filler loading, and cure/process conditions.
- Across the set, emulsion processing dominates the synthesis-relevant disclosures, while metathesis and compounding patents emphasize solvent choice, catalyst loading, temperature, and downstream property tuning rather than monomer feed composition.
- No patent in the set provides a direct high-ACN NBR recipe with explicit acrylonitrile wt% feed ratio and polymerization kinetics; the closest synthesis guidance comes from controlled emulsion polymerization and controlled radical polymerization disclosures.

4. CONCLUSION

The evidence indicates that the most practical synthesis route for high-acrylonitrile NBR is controlled emulsion copolymerization of acrylonitrile with butadiene, with precise monomer-feed

control and aqueous process management. US7592409B2 provides the strongest direct process framework for acrylonitrile-containing block structures, while EP3327047A1 broadens the controlled-polymerization toolbox to RAFT-based emulsion systems that explicitly include acrylonitrile monomers. EP2473279B1 and EP2057205B1 are valuable for post-synthesis molecular-weight and architecture tuning of nitrile rubbers, but they do not define the primary synthesis of high-ACN grades. Based on the disclosed evidence, recommended synthesis parameters are: emulsion polymerization in water, controlled radical or controlled emulsion conditions, careful monomer-ratio management, and downstream molecular-weight adjustment only if needed. For application development, the formulation patents show that nitrile rubbers are compatible with fillers, nanofillers, and reactive matrices, supporting use in high-performance elastomer composites.

5. REFERENCES

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- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- EP2561011B1 | Thermoplastic and/or elastomeric composite material based on carbon nanotubes ... | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2561011B1/en>
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